

Fullstack Developer | Machine Learning | Web Developer

Rayan Luqman Hakim

Informatics Portfolio 2026

About Me

I'am Rayan Luqman Hakim

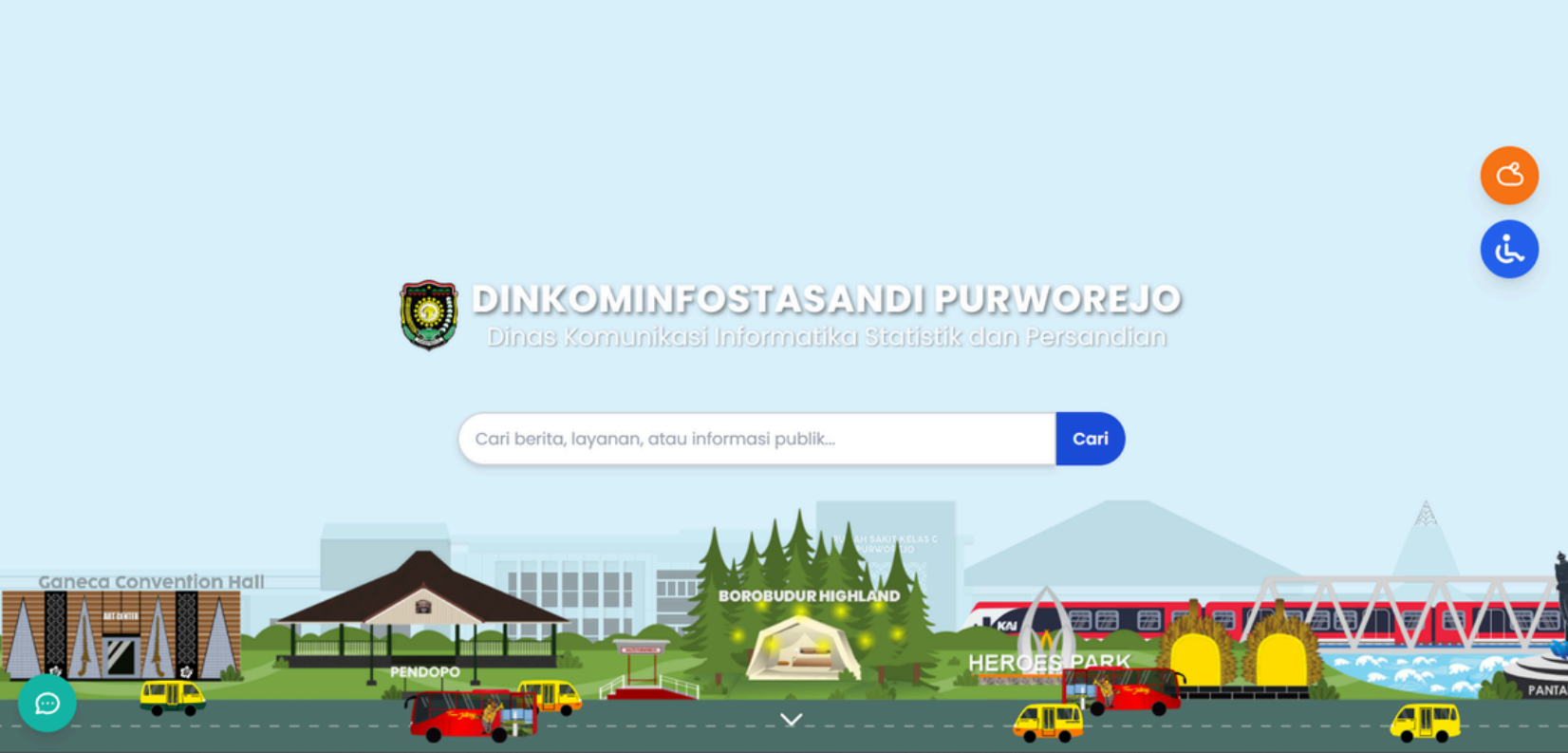
Informatics Engineering graduate specializing in Fullstack Development and Natural Language Processing (NLP), with experience in building web and mobile applications. Skilled in developing scalable systems using modern frameworks and applying machine learning techniques for data analysis and classification tasks



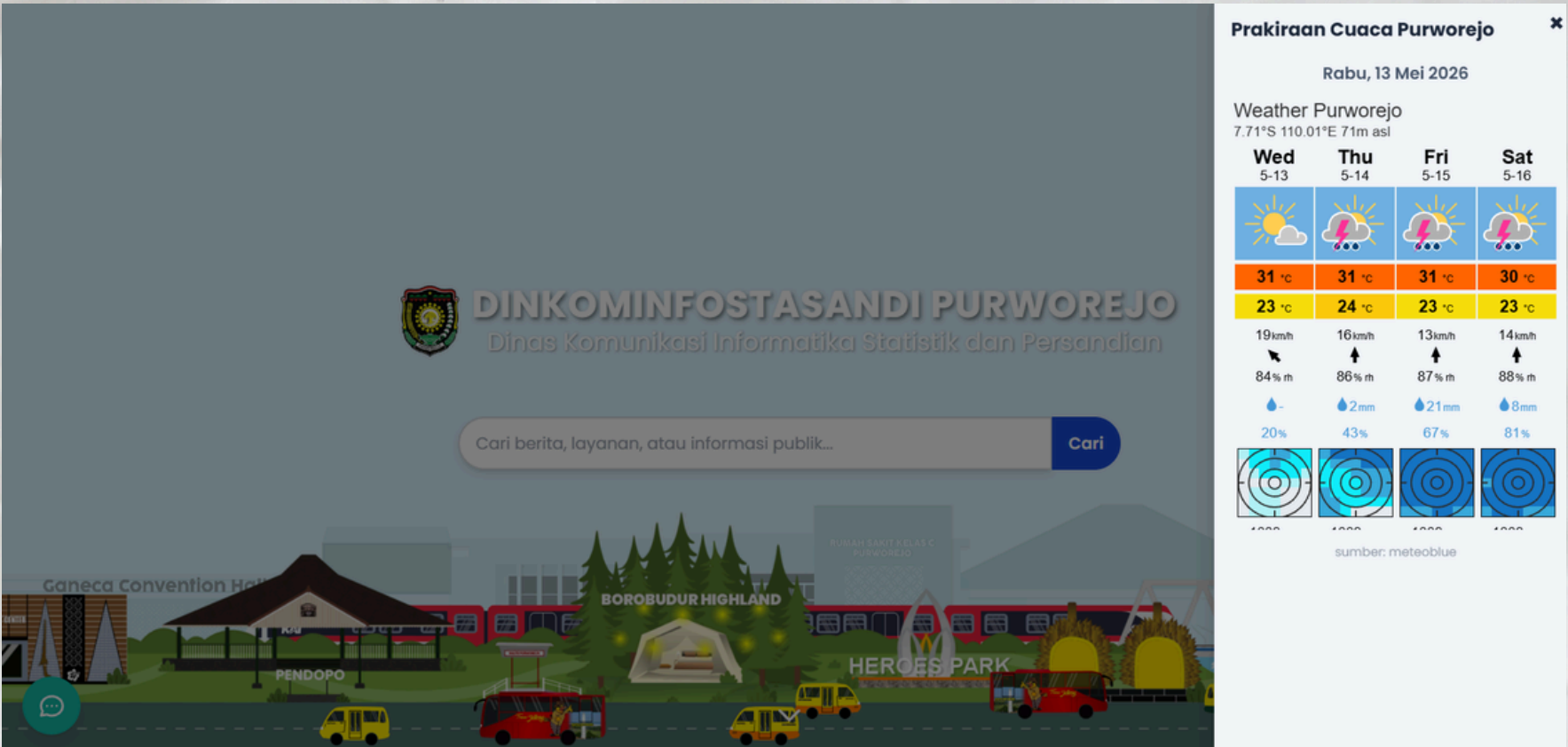
Website CMS Kominfo Purworejo

During my internship at Dinkominfo Sastandi Kabupaten Purworejo, I was challenged to bridge the communication gap between the local government and the community through a unified public information web portal. Built with Python (Flask), MySQL, and Jinja2, I developed a monolithic CMS system with a responsive frontend using HTML5, CSS3, Vanilla JavaScript, and Bootstrap 5.

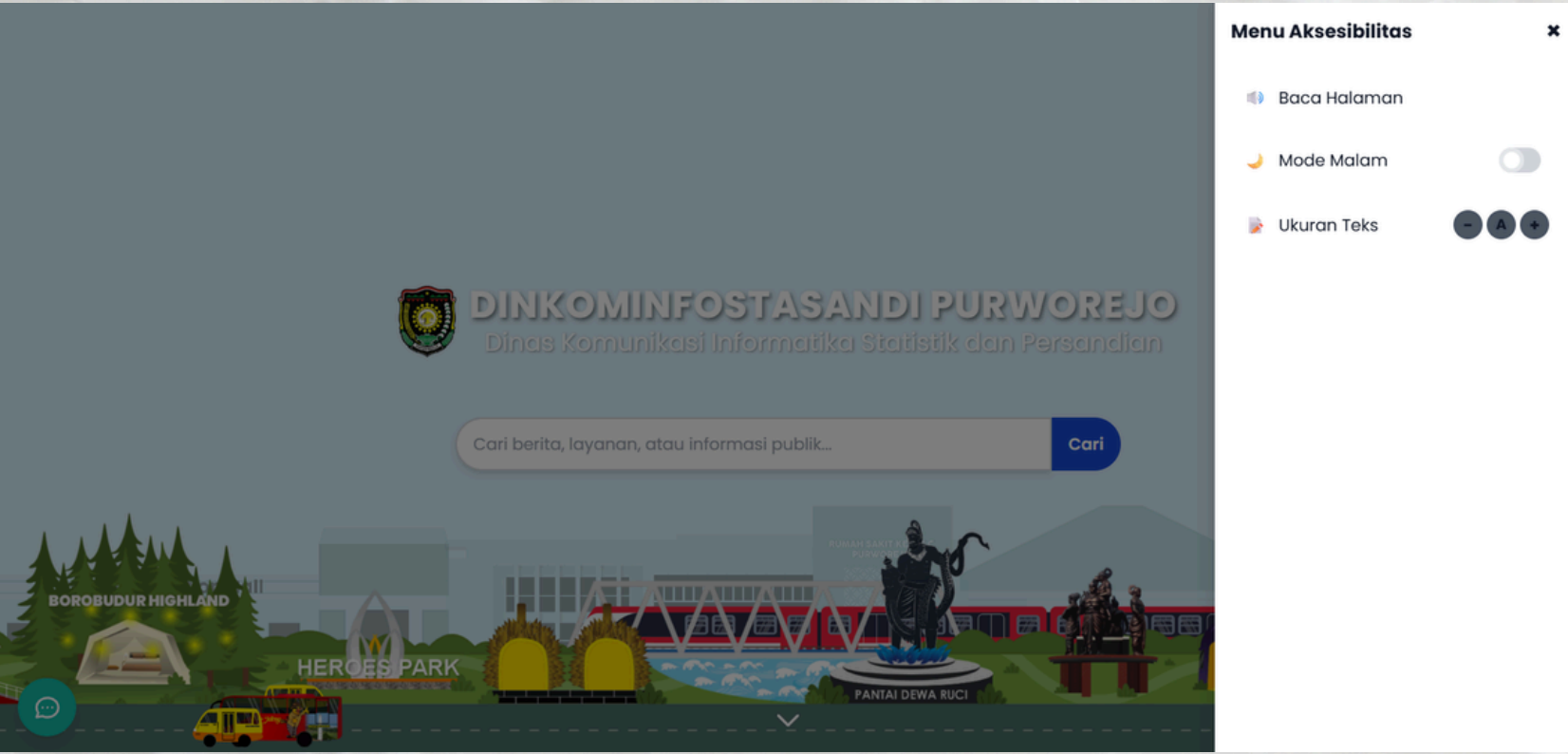
To make the platform more inclusive and user-friendly, I integrated the Web Speech API for Text-to-Speech accessibility, implemented Dark Mode, and developed a real-time Hoax Checker feature to help combat misinformation. From handling data integration with Newtonsoft.Json to building end-to-end system functionality, this project strengthened my skills in full-stack development, system analysis, and user-centered design.



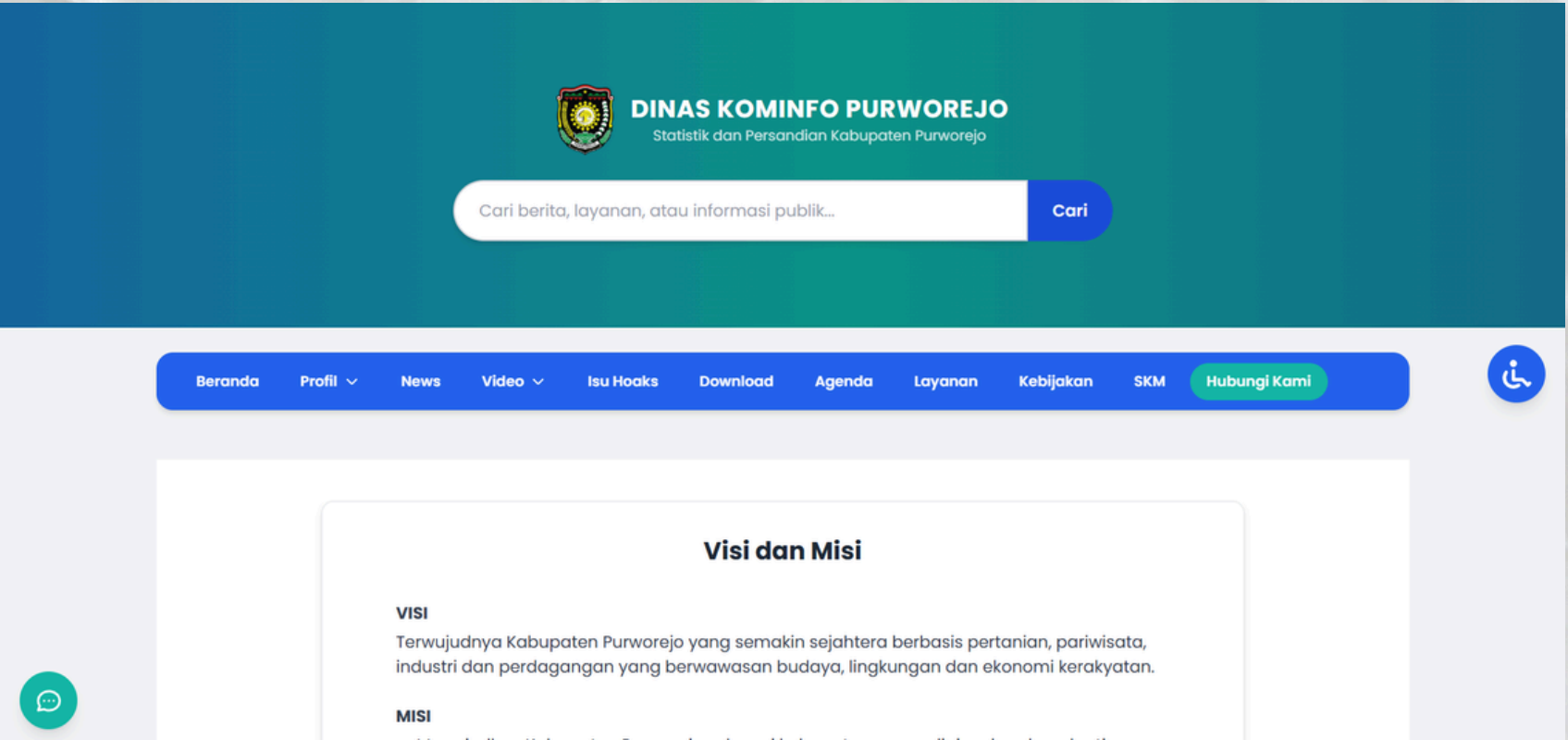
Halaman Utama



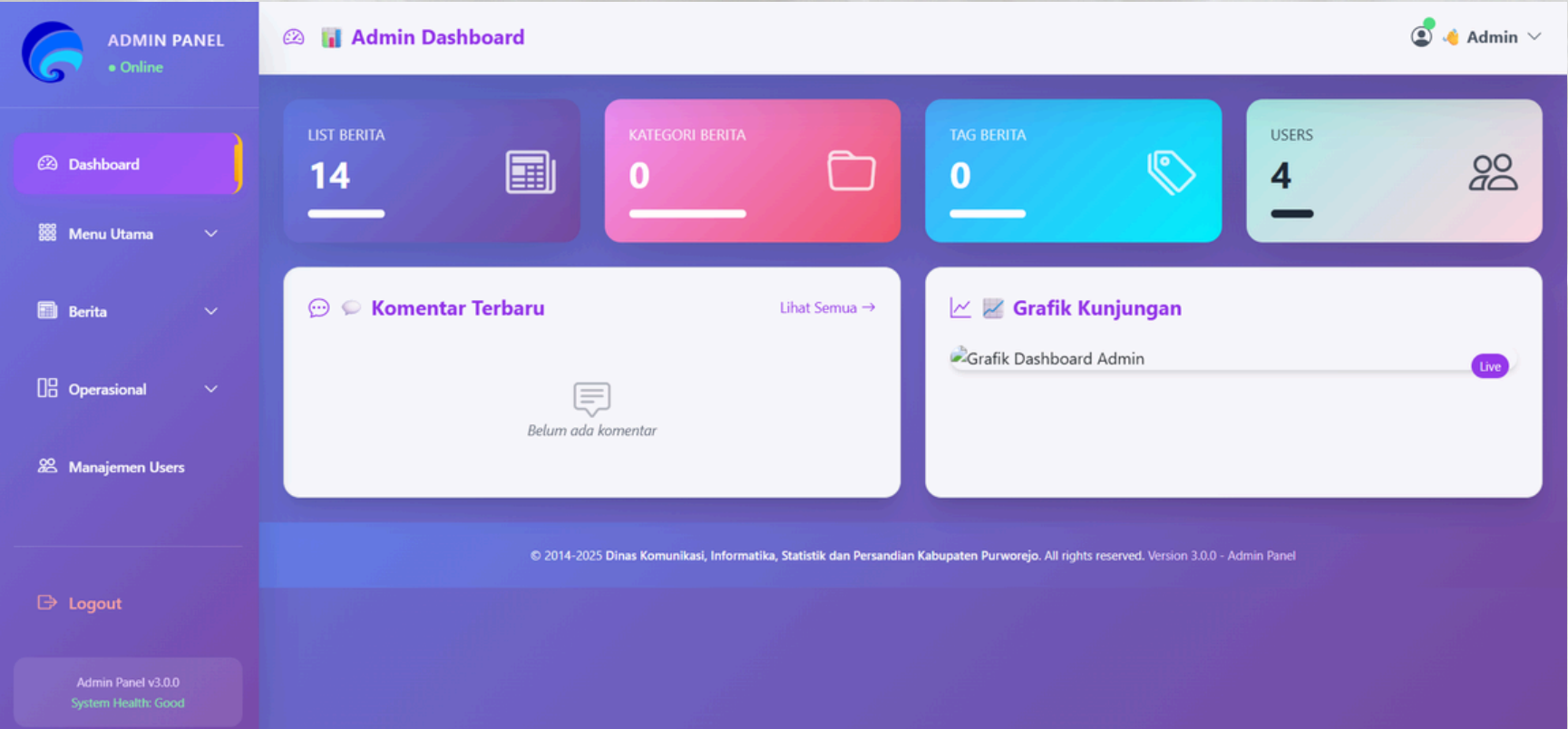
Fitur Cuaca



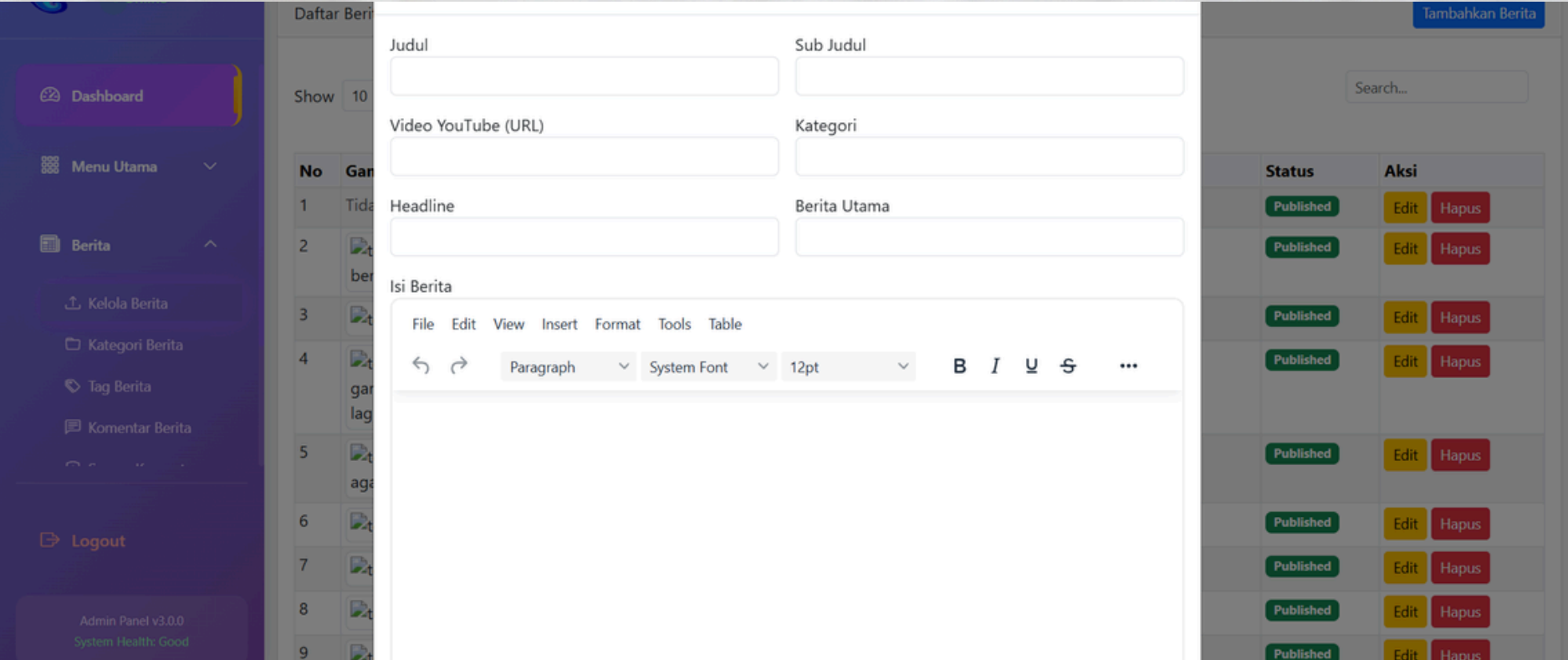
Menu Aksesibilitas



Halaman Visi



Dashboard Admin



Kelola Berita

Link Repository :
https://github.com/RayanHakim/DINKOMINFOSTASANDI_PURWOREJO_MAGANG

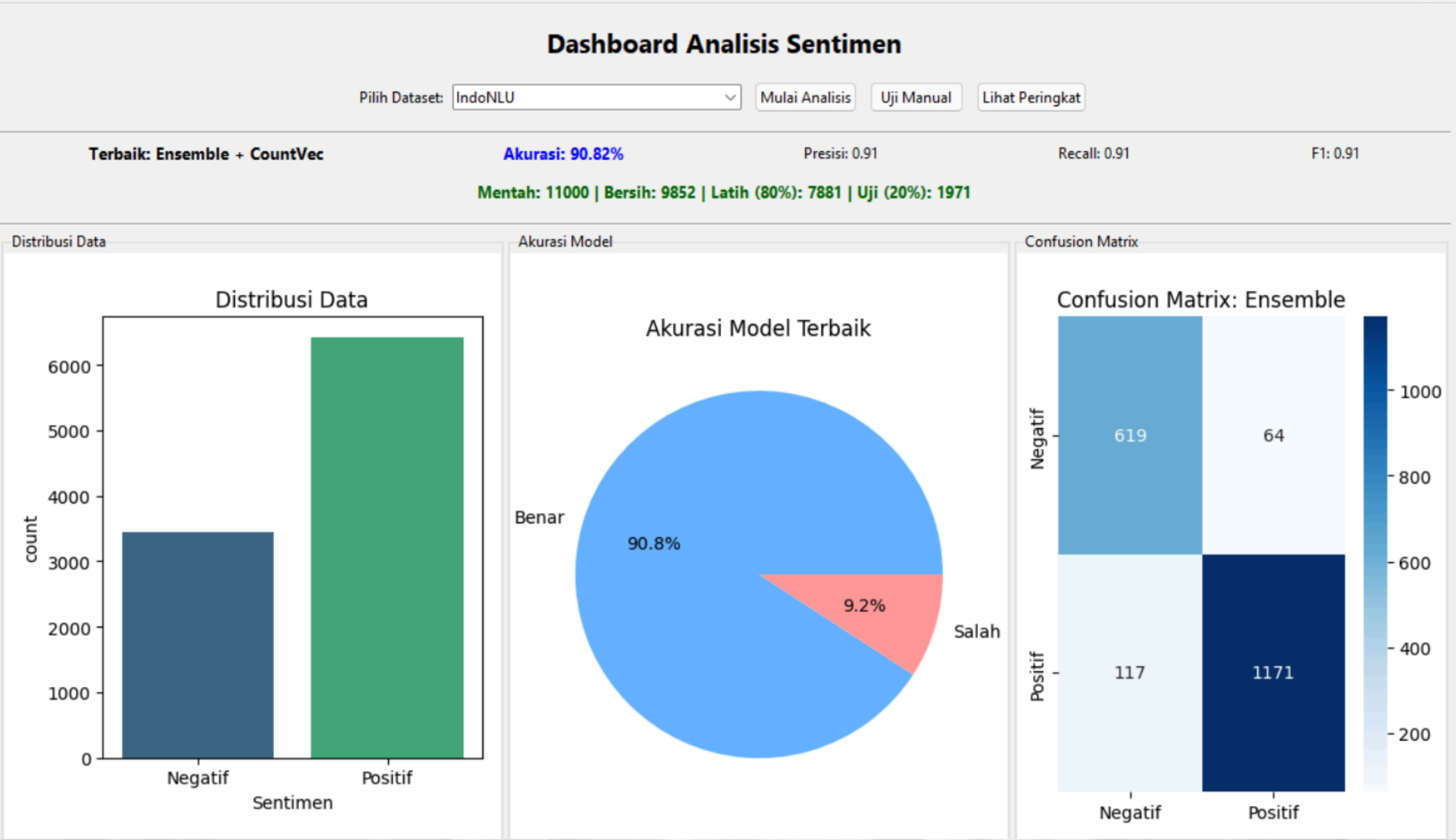
Sentiment Analysis Dashboard

For my undergraduate thesis, I developed a Sentiment Analysis Dashboard to evaluate public perception of electric vehicles. I personally performed manual sentiment labeling on the YouTube dataset to ensure high-quality ground truth, then built a Hybrid Ensemble Model in Python by combining Multinomial Naive Bayes and Logistic Regression via Soft Voting

I engineered a specialized NLP pipeline featuring Slang Normalization, Emoji Intelligence, and Negation Handling to tackle the nuances of Indonesian social media. Using Scikit-Learn and NLTK, I implemented SMOTE for class balancing and compared TF-IDF vs CountVectorizer to optimize the model's accuracy.

The final application, built with a Tkinter GUI, provides real-time visualizations using Matplotlib and Seaborn. This project successfully bridges academic research with functional software, transforming raw social media discourse into validated, interactive data insights.

Dashboard Analisis Sentimen



Hasil Akurasi

Peringkat Model				
Model	Akurasi	Presisi	Recall	F1-Score
Ensemble + CountVec	90.82%	0.91	0.91	0.91
Ensemble + TF-IDF	90.72%	0.91	0.91	0.91
LR + TF-IDF	90.26%	0.91	0.90	0.90
LR + CountVec	90.21%	0.91	0.90	0.90
NB + TF-IDF	89.45%	0.90	0.89	0.89
NB + CountVec	88.74%	0.89	0.89	0.89

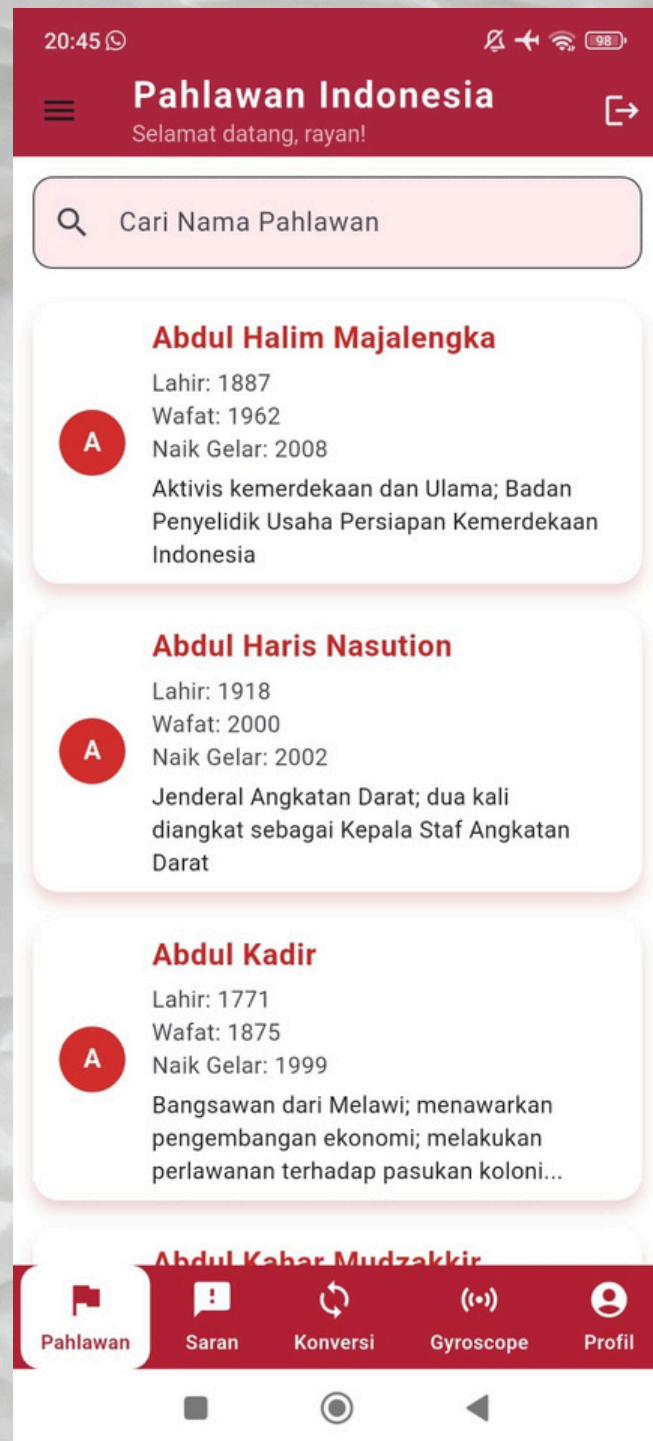
Link Repository :
<https://github.com/RayanHakim/Skripsi-Analisis-Sentimen-Mobil-Listrik-Cina>

Pahlawan Indonesia Mobile

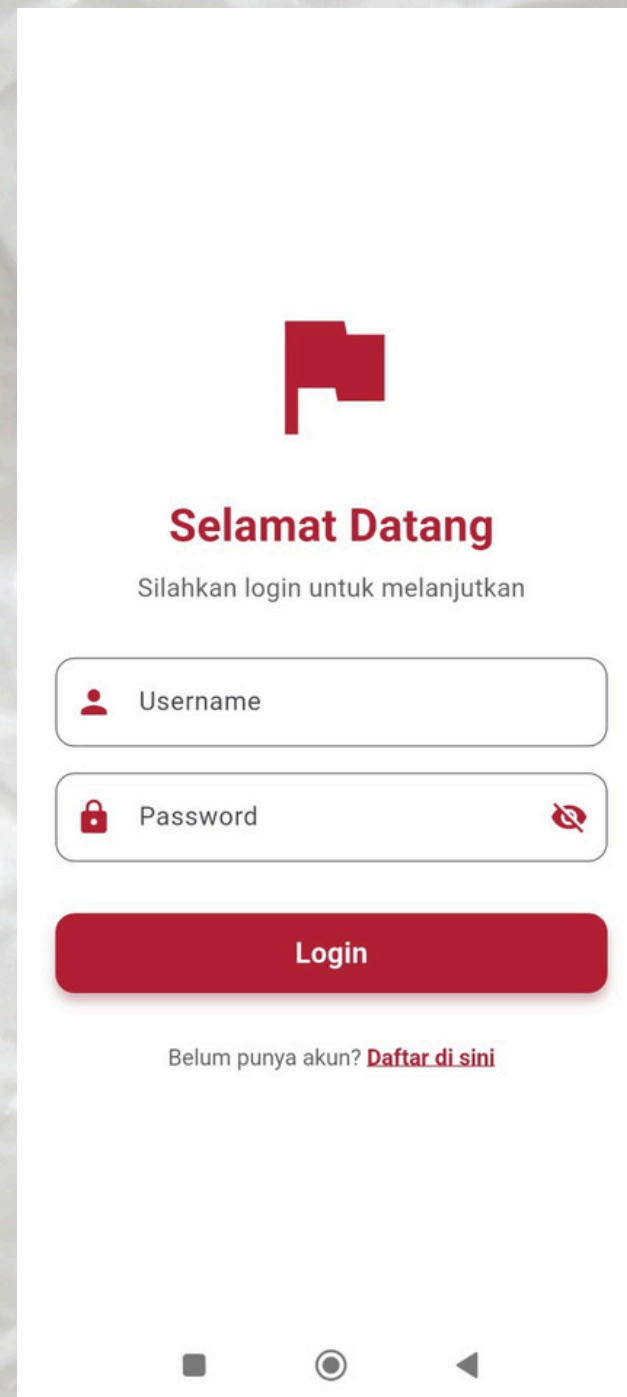
In this university project, I developed an educational mobile application called  Pahlawan Indonesia using Flutter and Dart. The application was designed to provide interactive information about Indonesian national heroes while implementing fundamental Android development concepts such as authentication, API integration, local databases, and device hardware access.

The application uses Sqflite for local database management, Shared Preferences for local storage, and HTTP for dynamic API integration. I also implemented features such as encrypted Login & Register systems using Crypto, gyroscope-based interactions with Sensors Plus, profile photo uploads using Image Picker, and local reminder notifications with Flutter Local Notifications.

Through this project, I improved my understanding of mobile application architecture, state management with FutureBuilder and StatefulWidget, device permission handling, and user-focused feature development in Flutter.



Halaman Utama



Halaman Login



Sensor Gyroscope

Link Repository : https://github.com/RayanHakim/Project_Akhir_Mobile

Flagfall Arena

Flagfall Arena is a 2D MOBA-inspired Capture the Flag game that I developed using Python and Pygame. The project focuses on combining fast-paced gameplay mechanics with strategic team-based combat, allowing players to choose different character classes and compete across multiple fantasy-themed arenas.

The game features multiple playable classes including Assassin, Knight, Tank, Wizard, and Support, each with unique abilities, combat styles, and special skills. I also implemented dynamic camera systems, AI-controlled bots, character-specific projectiles, respawn mechanics, and immersive audio systems with different background music and sound effects for gameplay events.

Through this project, I explored game development concepts such as real-time gameplay systems, AI behavior, collision handling, animation flow, and object-oriented programming in Python using the Pygame library.

Flagfall Arena

Back

CHARACTER INDEX

< [LEFT]



[RIGHT] >

THE ASSASSIN

HP: 70 | ATK: 25 | SPD: 8.0

Seorang pembunuh bayaran yang bergerak dalam bayang-bayang. Memiliki kecepatan luar biasa namun pertahanan yang sangat rapuh.

ABILITIES:

[F] Shadow Slash: Tebasan pedang super cepat.

[G] Throwing Knife: Serangan jarak jauh ringan.

[H] Stealth: Menghilang sejenak dari pandangan.

[J] Shadow Dash: Berpindah tempat secara instan.

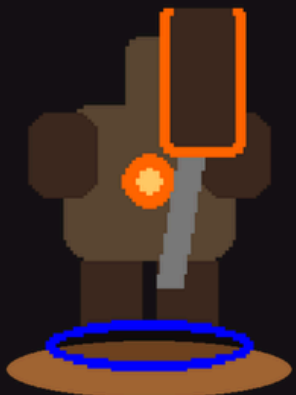
Hero 1 of 5 | Press [BACK] to Menu

Flagfall Arena

Back

CHARACTER INDEX

< [LEFT]



[RIGHT] >

THE TANK

HP: 300 | ATK: 10 | SPD: 2.5

Benteng berjalan yang hampir tidak bisa dihancurkan. Menggunakan palu godam untuk menggetarkan bumi.

ABILITIES:

[F] Hammer Slam: Menghantam tanah di depan.

[G] Rock Throw: Melempar batu besar ke musuh.

[H] Fortify: Menjadi tak tergoyahkan (Immune).

[J] Earthshatter: Shockwave masif di area luas.

Hero 3 of 5 | Press [BACK] to Menu

Flagfall Arena

Back

CHARACTER INDEX

< [LEFT]



[RIGHT] >

THE SUPPORT

HP: 100 | ATK: 8 | SPD: 5.0

Penyembuh mistis yang fokus utama menjaga kawan tetap hidup dengan aura penyembuhan suci.

ABILITIES:

[F] Staff Poke: Serangan ringan jarak dekat.

[G] Light Orb: Menembakkan bola cahaya suci.

[H] Protective Aura: Membuat tameng energi tim.

[J] Area Healing: Memulihkan HP diri dan kawan.

Hero 5 of 5 | Press [BACK] to Menu

Flagfall Arena

Back

CHARACTER INDEX

< [LEFT]



[RIGHT] >

THE KNIGHT

HP: 150 | ATK: 15 | SPD: 4.5

Ksatria kerajaan yang kokoh dengan zirah baja. Penyeimbang antara daya tahan dan kekuatan serangan di garda depan.

ABILITIES:

[F] Broadsword: Tebasan pedang berat.

[G] Shield Toss: Melempar perisai ke arah musuh.

[H] Shield Wall: Bertahan total dari damage.

[J] Brave Charge: Menerjang musuh dengan brutal.

Hero 2 of 5 | Press [BACK] to Menu

Flagfall Arena

Back

CHARACTER INDEX

< [LEFT]



[RIGHT] >

THE WIZARD

HP: 80 | ATK: 20 | SPD: 4.0

Penguasa elemen dari menara sihir. Menyerang dari jarak jauh dengan kekuatan sihir tingkat tinggi.

ABILITIES:

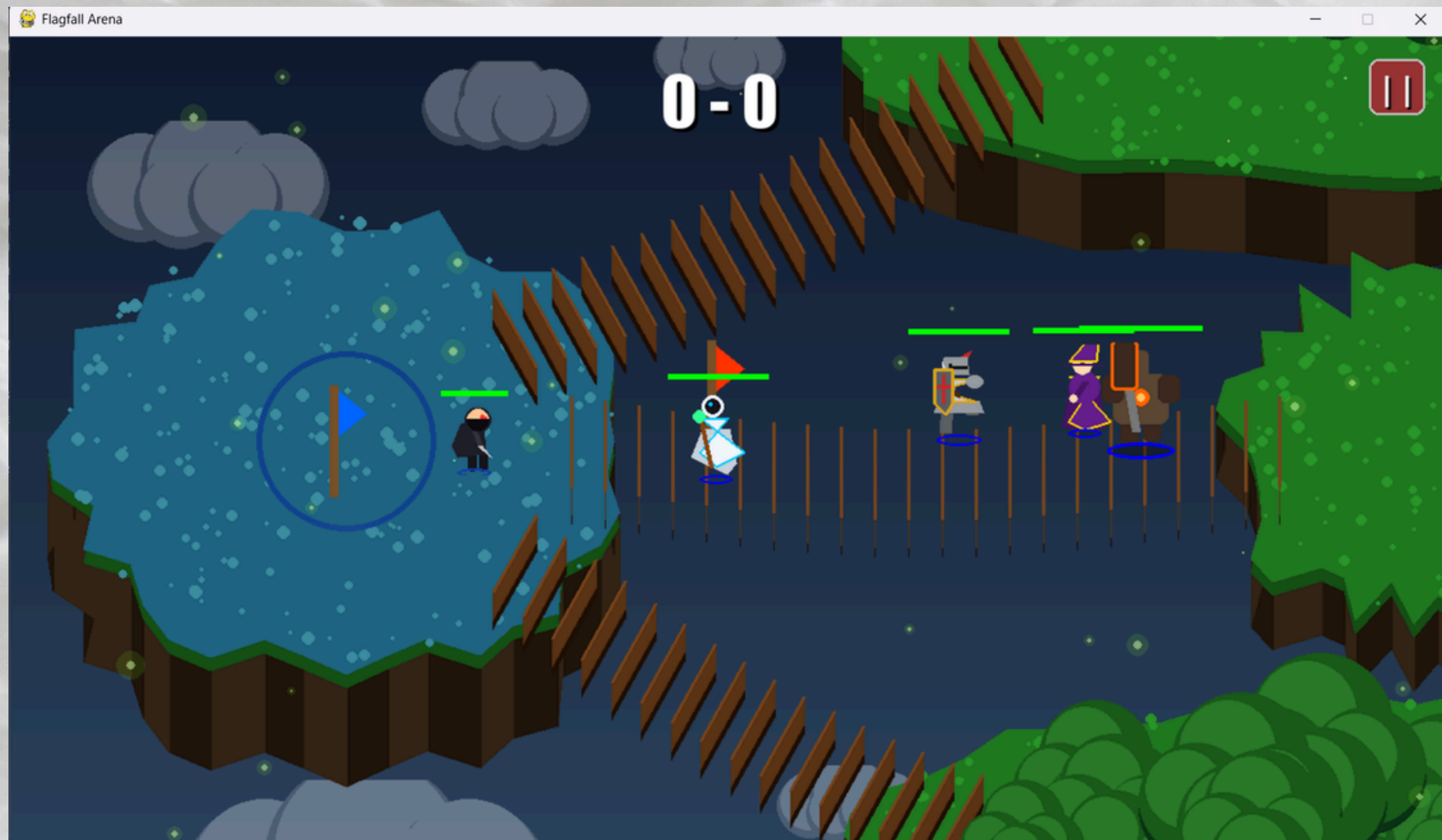
[F] Magic Staff: Pukulan tongkat jarak dekat.

[G] Fireball: Tembakan energi api beruntun.

[H] Blink: Teleportasi jarak pendek.

[J] Meteor Strike: Hujan api dahsyat dari langit.

Hero 4 of 5 | Press [BACK] to Menu



Link Repository : <https://github.com/RayanHakim/Flagfall-Arena>

Cafe Purworejo

Cafe Purworejo is a simple web platform designed to help local residents and tourists discover popular cafes in Purworejo. Developed as a front-end practice project using React and Vite, the website focuses on delivering a clean, responsive, and user-friendly browsing experience across desktop and mobile devices.

The platform features a categorized cafe directory with detailed information such as addresses, locations, and official Instagram links. I developed the application using React.js, Vite, and Tailwind CSS, while implementing responsive layouts and automatic alphabetical sorting to improve usability and navigation.

To streamline deployment and project maintenance, I integrated GitHub Actions for automated CI/CD workflows and deployed the website through [GitHub Pages](#). Through this project, I improved my understanding of modern front-end development and responsive web design practices.

CAFE PURWOREJO

Daftar Cafe Di Purworejo.

🔍 Cari nama cafe favoritmu...



Original Link Photo

Echosistem

📍 Jl. Arta Kencana No.6, RT.01/RW.16, Plaosan, Purworejo, Kec. Purworejo,



Original Link Photo

EXOTIC Coffee & Eatery

📍 Jl. Brigjen Katamso No.115, RT.01/RW.03, Rw. 03, Pangenrejo, Kec. Purworejo,



Original Link Photo

Kebon N'dalem

📍 Jl. Eyang Surogati, Krajan, Paduroso, Kec. Purworejo, Kabupaten Purworejo, Jawa

CAFE PURWOREJO

Daftar Cafe Di Purworejo.

🔍 Cari nama cafe favoritmu...

Original Link Photo

Kedai Pekarangan

📍 Jalan Semawung & Jl. Bagelen - Cangkep, Dusun III, Cangkreplor, Kec. Purworejo, Kabupaten Purworejo, Jawa Tengah 54117



Original Link Photo

Kopi Candi

📍 Jl. Dukuh Parakan No.rt. 01/4, Paraan, Candingasinan, Kec. Banyuurip, Kabupaten Purworejo, Jawa Tengah 54171



Original Link Photo

Kopi Kembang Sepasang

📍 Jl Brenggong No.58, Brenggong Kidul, Brenggong, Purworejo, Purworejo Regency, Central Java 54119



SINDURJAN



PANGENJURUTENGGAH



SINDURJAN

Link Repository : <https://github.com/RayanHakim/Cafe-Purworejo>